

EE/CS 172/130
Digital Logic and Design

Topic 5 – Standard Forms

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Standard (Canonical) Forms

- A logic expression can be written in a number of equivalent forms.
- Two such forms are:
 - Sum of product (SOP)
 - Example: $A\bar{B}C + \bar{A}B + \bar{A}\bar{B}\bar{C}$
 - Product of sums (POS)
 - Example: $(A + \bar{B} + C)(\bar{A} + B)(\bar{A} + \bar{B} + \bar{C})$

Standard SOP

- An SOP expression is standard SOP expression if every product term involves every variable or its complement.
- **Example:** $F(A, B, C) = \bar{A}\bar{B}C + ABC + \bar{A}\bar{B}\bar{C}$
- Each product term of standard expression is called a **minterm** (m).
- The standard expression may not be the simplest expression.
- A logic expression can directly be obtained from a truth table by writing one product term (minterm) for each input combination that produces an output of 1.
- If A is a Boolean variable in a minterm:
 - If $A = 1$, it is written as A .
 - If $A = 0$, it is written as $\bar{A}$ (A complemented).

<i>A</i>	<i>B</i>	<i>C</i>	Minterm, <i>m</i>
0	0	0	$\bar{A}\bar{B}\bar{C}, m_0$
0	0	1	$\bar{A}\bar{B}C, m_1$
0	1	0	$\bar{A}B\bar{C}, m_2$
0	1	1	$\bar{A}BC, m_3$
1	0	0	$A\bar{B}\bar{C}, m_4$
1	0	1	$A\bar{B}C, m_5$
1	1	0	$AB\bar{C}, m_6$
1	1	1	ABC, m_7

Standard POS

- A POS expression is standard POS expression if every sum term involves every variable or its complement.
- **Example:** $F(A, B, C) = (A + \bar{B} + C)(A + B + C)(\bar{A} + \bar{B} + \bar{C})$
- Each sum term of standard expression is called a **maxterm** (M).
- The standard expression may not be the simplest expression.

- A logic expression can directly be obtained from a truth table by writing one sum term (maxterm) for each input combination that produces an output of 0.
- If A is a Boolean variable in a maxterm:
 - If $A = 0$, it is written as A .
 - If $A = 1$, it is written as $\bar{A}$ (A complemented).

A	B	C	Maxterm, M
0	0	0	$(A + B + C), M_0$
0	0	1	$(A + B + \bar{C}), M_1$
0	1	0	$(A + \bar{B} + C), M_2$
0	1	1	$(A + \bar{B} + \bar{C}), M_3$
1	0	0	$(\bar{A} + B + C), M_4$
1	0	1	$(\bar{A} + B + \bar{C}), M_5$
1	1	0	$(\bar{A} + \bar{B} + C), M_6$
1	1	1	$(\bar{A} + \bar{B} + \bar{C}), M_7$

Example

- SOP expression

$$F(A, B, C) = \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$$

- POS expression

$$F(A, B, C) = (A + B + C)(A + \bar{B} + \bar{C})(\bar{A} + B + \bar{C})(\bar{A} + \bar{B} + C)$$

- Both expressions are equivalent.

- **Proof:**

$$\begin{aligned}\text{POS} &= (A + B + C)(A + \bar{B} + \bar{C})(\bar{A} + B + \bar{C})(\bar{A} + \bar{B} + C) \\ &= [A + (B + C)(\bar{B} + \bar{C})][\bar{A} + (B + \bar{C})(\bar{B} + C)] \\ &= [A + B\bar{B} + B\bar{C} + \bar{B}C + C\bar{C}][\bar{A} + B\bar{B} + BC + \bar{B}\bar{C} + C\bar{C}] \\ &= [A + 0 + B\bar{C} + \bar{B}C + 0][\bar{A} + 0 + BC + \bar{B}\bar{C} + 0] \\ &= [A + B\bar{C} + \bar{B}C][\bar{A} + BC + \bar{B}\bar{C}] \\ &= A\bar{A} + ABC + A\bar{B}\bar{C} + \bar{A}B\bar{C} + BBC\bar{C} + B\bar{B}\bar{C}\bar{C} + \bar{A}\bar{B}C + B\bar{B}CC + \bar{B}\bar{B}C\bar{C} \\ &= 0 + ABC + A\bar{B}\bar{C} + \bar{A}B\bar{C} + 0 + 0 + \bar{A}\bar{B}C + 0 + 0 \\ &= ABC + A\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}\bar{B}C = \text{SOP}\end{aligned}$$

<i>A</i>	<i>B</i>	<i>C</i>	<i>F</i>
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

Alternate Form

SOP expression

- $F(\text{input variables}) = \sum(\text{list of minterms})$

Example:

$$\begin{aligned} F(A, B, C) &= \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC \\ &= \sum(1, 2, 4, 7) \end{aligned}$$

POS expression

- $F(\text{input variables}) = \pi(\text{list of maxterms})$

Example:

$$\begin{aligned} F(A, B, C) &= (A + B + C)(A + \bar{B} + \bar{C})(\bar{A} + B + \bar{C})(\bar{A} + \bar{B} + C) \\ &= \pi(0, 3, 5, 6) \end{aligned}$$

- If output has lesser number of 1s, use SOP.
- If output has lesser number of 0s, use POS.

<i>A</i>	<i>B</i>	<i>C</i>	<i>F</i>
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

Converting to Standard SOP

- AND the missing variable in the form $(x + \bar{x})$.

Example

- $$\begin{aligned} F(A, B, C) &= AB + A\bar{B}C + \bar{B} \\ &= AB(C + \bar{C}) + A\bar{B}C + \bar{B}(A + \bar{A})(C + \bar{C}) \\ &= ABC + AB\bar{C} + A\bar{B}C + (\bar{A}\bar{B} + A\bar{B})(C + \bar{C}) \\ &= ABC + AB\bar{C} + A\bar{B}C + \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C} + A\bar{B}C + A\bar{B}\bar{C} \\ &= ABC + AB\bar{C} + A\bar{B}C + \bar{A}\bar{B}\bar{C} + A\bar{B}C + \bar{A}\bar{B}C \\ &= \Sigma(0,1,4,5,6,7) \end{aligned}$$

Converting to Standard POS

- OR the missing variable in the form $(x\bar{x})$.

Example

- $$\begin{aligned} F(A, B, C) &= (A + B)(A + \bar{B} + C)(\bar{B}) \\ &= (A + B + C\bar{C})(A + \bar{B} + C)(\bar{B} + A\bar{A}) \\ &= (A + B + C)(A + B + \bar{C})(A + \bar{B} + C)(A + \bar{B})(\bar{A} + \bar{B}) \\ &= (A + B + C)(A + B + \bar{C})(A + \bar{B} + C)(A + \bar{B} + C\bar{C})(\bar{A} + \bar{B} + C\bar{C}) \\ &= (A + B + C)(A + B + \bar{C})(A + \bar{B} + C)(A + \bar{B} + C)(A + \bar{B} + \bar{C})(\bar{A} + \bar{B} + C)(\bar{A} + \bar{B} + \bar{C}) \\ &= (A + B + C)(A + B + \bar{C})(A + \bar{B} + C)(A + \bar{B} + \bar{C})(\bar{A} + \bar{B} + C)(\bar{A} + \bar{B} + \bar{C}) \\ &= \pi(0, 1, 2, 3, 6, 7) \end{aligned}$$

Problems

- Design a 3-bit prime detector.

Solution

- Let A , B , and C be the three variables representing the input number.
- $$\begin{aligned} F(A, B, C) &= \sum(2, 3, 5, 7) \\ &= \bar{A}\bar{B}\bar{C} + \bar{A}BC + A\bar{B}\bar{C} + ABC \\ &= \bar{A}B(\bar{C} + C) + AC(\bar{B} + B) \\ &= \bar{A}B(1) + AC(1) \\ &= \bar{A}B + AC \end{aligned}$$
- Draw circuit diagram.

Problems

- Design a logic circuit whose output is 1 if the 3-bit input value is completely divisible by 2 or 5.
[Answer: $B\bar{C} + A\bar{B}$]
- Design a logic circuit whose output is 1 if the 3-bit input value is in the power of 2.
[Answer: $\bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C}$]
- Design a logic circuit whose output is 1 if majority of the inputs is 1. Consider a 3-bit input.
[Answer: $\bar{B}C + A\bar{C} + A\bar{B}$]
- Design a logic circuit whose output is 1 if majority of the inputs is 1. Consider a 4-bit input.